

Stefanos Englezou

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PROFILE

I am a Computer Science graduate from the University of Cyprus, currently pursuing an MSc in Informatics at the University of Zurich. My research interests lie at the intersection of deep learning, bioinformatics and computational neuroscience, with a particular focus on applying neural network architectures to problems in these fields, such as protein structure prediction and self-control modelling. I am also exploring the applications of spiking neural networks and neuromorphic computing as biologically plausible alternatives to conventional artificial neural networks.

EDUCATION

University of Zurich, Zurich, Switzerland

09/2026

MSc in Informatics, *Major in Artificial Intelligence / Minor in Bioinformatics*

University of Cyprus, Nicosia, Cyprus

09/2022 – 06/2026

BSc in Computer Science

Graduated with GPA: 8.37/10

Thesis: *“Protein Secondary Structure Prediction with machine learning techniques”*

Advisor: Prof. Chris Christodoulou

- Designed and implemented an ensemble deep learning pipeline for Protein Secondary Structure Prediction (PSSP), combining an encoder-only Transformer with frozen ESM-2 protein language model embeddings for both 3-state and 8-state classification tasks achieving comparable results with state-of-the-art methods on benchmarks such as CASP12, CB513 and TS115.
- Developed a post-processing module consisting of a filtering algorithm (Random Forest) and empirical rules for unfeasible segments, to refine per-residue predictions, improving segment-level SOV scores and minority class prediction.
- Conducted interpretability analysis of learned ESM-2 representations via PCA and attention weight visualization, confirming correct model behavior and class separability across secondary structure states.

WORK EXPERIENCE

Software Developer Intern

06/2025 - 08/2025

Grant Thornton Cyprus – Quantitative Risk Department, Nicosia, Cyprus

- Built an internal CV-Proposal System using the Microsoft Power Platform with vector databases to digitize employee and project data.
- Implemented a hybrid employee matching engine: TF-IDF inverted index for lexical ranking, re-ranked by semantic similarity via OpenAI text embeddings and Large Language Model (LLM) integration.
- Automated proposal generation by compiling ranked employee data into structured PowerPoint presentations, enabling data-driven team selection.

Research Engineer

01/2024 - 01/2026

KIOS Center of Excellence -- Water Distribution Team, Nicosia, Cyprus

- Developed QGIS plugins in Python with Qt Designer; built and optimised scalable Python applications (testing and debugging, refactoring, documentation).
- Contributed to the Oceanos Digital Twin (OceanosDT): water-network monitoring, model calibration, and genetic algorithm simulations for the Water Boards of Limassol & Larnaca.
- Lead developer of WBLSampling Django web app: geospatial data management with Leaflet.js maps, jQuery-driven DB queries, and a PostgreSQL back-end.

TECHNICAL

- ✓ **Machine Learning & Deep Learning:** Familiar with classification, clustering, and regression techniques. Experienced with neural network architectures (RNNs, CNNs and Transformers) with applications across Natural Language Processing, Computer Vision and Bioinformatics. Technologies known: PyTorch, Scikit-learn, Pandas, NumPy, Optuna, OpenCV and Hugging Face Transformers.
- ✓ **Computational Neuroscience:** Familiar with neuron models (e.g. LIF, HH), including integration of synaptic, spatial and ionic mechanisms, simulation using NEURON and Brian, exposure to neural coding, spiking neural network models and learning rules (Hebbian learning, STDP).

AWARDS & PRIZES

- Aridemi Foundation Scholarship – € 3000 08/2026
- First place of team project for Databases course, University of Cyprus 12/2024